

THE INTERNET OF THINGS



News, views and development on the Internet of Things

“Bluetooth Smart offers rapid rollout of low-power consumer products around the latest smartphone and tablet platforms, creating a large addressable market”

Bluetooth SIG has announced the Bluetooth Core Specification 4.1, which they say improves usability for consumers, empowers innovation for product developers and extends the technology's foundation as an essential link for the IoT. This is therefore a good time to talk to Nitin Gupta, lead engineer, μ Energy Applications, CSR India Pvt Ltd, an expert in Bluetooth Smart, about the relevance of the low-power communication technology for the IoT era

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The IoT requires almost all objects to have some communication ability built into them. What is the scope of Bluetooth Smart in the IoT scheme of things?

Smart devices, both tablets and smartphones, are likely to be central to the IoT. For direct consumer control and gathering data from low-power devices, Bluetooth Smart is ideal. With developer APIs available on all the major OS platforms, Bluetooth Smart may provide the first wave of IoT devices.

Bluetooth is likely to work with Wi-Fi to deliver consumer IoT products due to the huge deployment rates of these technologies.

What are the other options for M2M communication, and what are the advantages of Bluetooth Smart over these?

For M2M, cellular communication options are available, but these attract a service cost which can be prohibitive for products, as well as higher power consumption. These are typically much higher cost per unit, and also much

higher power consumption. Looking forward, whitespace radios using the weightless standard may have a role to play where longer range is required in monitoring infrastructure.

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It is said that Bluetooth Smart is a completely new technology as compared to classical Bluetooth. Is that so? If so, how does an engineer decide which one best suits his/her needs?

Bluetooth Smart introduces significant changes to the radio for the low-energy peripherals. It is optimised for applications that transfer small amounts of data. Peak throughput is limited to 120kbps. Bluetooth Classic is optimised for streaming data in the range of 300kbps-1Mbps, making it perfect for audio or high-data-rate sensor applications.

What are the main shortcomings of Bluetooth Smart 4.0, and which of these

have been addressed in the recently announced 4.1 specification?

There are no mandatory changes in the 4.1 specification but it does open up some further flexibility. The Bluetooth Smart 4.0 version of the spec enables star topology networks only. This means that a smart watch can only be a master to a set of slave sensors or a slave to a smartphone, but not both master and slave at the same time. For developers, this means that the smart watch has to switch roles to support both connections. In the 4.1 specification, it is possible for the watch to be a slave to the smartphone whilst maintaining its position as a master to several slave sensors.

Bluetooth Smart support in Android is expected to cause a boom in IoT objects—what is your take on this?

Bluetooth Smart support in Android is a huge step, and Android continues to dominate market share in smart devices. I therefore agree that support in Android will enable new innovation and encourage more developers to work with the technology.